Purdue University CHE 20500: Chemical Engineering Calculations, Spring 2026

A. Instructor. Professor Tayler Hebner (thebner@purdue.edu)

B. Catalog Description. Quantitative applications of steady-state mass and energy balances to solve problems involving multi-component systems and multi-unit chemical processes. Single-component and multi-component phase equilibria, single-reaction and multiple-reaction stoichiometry, coupled mass and energy balances, chemical processes involving bypass and recycle streams.

C. Prerequisites. Chemistry 116 or Chemistry 136; Mathematics 161 (or equivalent); Physics 172 (or equivalent)

D. Course Text. *Elementary Principles of Chemical Processes, 4th Edition* **with the WileyPlus account** by R. M. Felder, R. W. Rousseau, and L. G. Bullard. Wiley and Sons, 2016. Purchases can be made through the bookstore or directly from Wiley.

- WileyPLUS Access with downloadable eText ISBN: 9781119498704
- WileyPLUS Access with downloadable eText and Loose Leaf 1 semester ISBN: 9781119760818

E. Course Learning Objectives. It is expected that, by the conclusion of the semester, class participants should be able to:

- Work professionally and ethically as a member of a chemical engineering team.
- State and describe the diverse social, economic, and environmental issues associated with being a chemical engineer.
- Apply the law of conservation of mass and conservation of atomic species in order to solve mass balances in unit operations with and without chemical reactions and with and without recycle streams.
- Determine, using first principles and well-established correlations, the relations between thermodynamic equilibria and multiphase systems.
- Integrate the first law of thermodynamics with the concept of energy balances in unit operations with and without chemical reactions and with and without recycle streams.
- Apply the laws of conservation of mass and energy and thermodynamic equilibrium data in order to formulate solutions for mass and energy flow rates in multi-unit systems.
- Utilize the concepts of transient mass balance problems in order to develop a basis for non-steady state applications.
- Design multi-unit chemical processes using steady-state and transient mass and energy balances in order to create multi-unit operations similar to those in future courses and applications in industry.

F. Course Outcomes. By the end of the course, the student should be able to:

1. Estimate physical properties of real chemical systems (Utilized in CHE 21100, 30600, 34800, 37700, 37800, 42000, 43500, 45000)
2. Evaluate introductory single-component and multi-component phase equilibria and incorporate these concepts into solutions of mass and energy balance problems (Utilized in CHE 21100, 30600, 37800, 43500, 45000)
3. Solve steady state and transient mass and energy balance problems for both reacting and non-reacting systems with or without recycle using analytical and computational methods (Utilized in CHE 21100, 30600, 34800, 37700, 37800, 42000, 43500, 45000, 45600)
4. Work professionally and ethically in teams to solve mass and energy balance problems (Utilized in CHE 30600, 34800, 37700, 37800, 43500, 45000)
5. Identify contemporary chemical engineering problems, including their impact on societal, economic, public welfare, environmental, and global factors (Utilized in CHE 30000, 40000, 42000, 45000)

G. **Davidson School of Chemical Engineering Program Outcomes for ABET.** Graduates of the Charles D. Davidson School of Chemical Engineering at Purdue University will (**bolded items are addressed in this course**):

1. **Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems.**
2. **Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.**
3. Communicate effectively with a range of audiences.
4. **Recognize ethical and professional responsibilities in chemical engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.**
5. **Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.**
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. **Acquire and apply new knowledge as needed, using appropriate learning strategies.**

H. **Expectations.** This is a 4 credit hour course, and it is expected that each student will spend a minimum of 12 hours each week (3 hours per credit) outside of lectures working on homework, studying, reading the course text, and completing assessment questions. In many cases, the first year in the university is harder than high school, but it is manageable with noticeably less effort than described above. This class, and all subsequent chemical engineering classes, will be much more challenging, and hence more rewarding, than anything that most students will have seen in previous courses. We encourage you to take studying seriously and establish good study habits (e.g., read the text during the assigned week, practice using additional problems). This will lead to a successful start of the student's chemical engineering career.

I. **Instructor's Commitment.** Your instructor will: (1) be courteous, punctual, well-organized, and prepared for class activities; (2) answer questions clearly in class or through office hours; (3) be available during office hours or notify you beforehand if they are unable to keep to the original office hour schedule; and (4) grade uniformly and consistently to the posted guidelines.

J. **Consulting with the Faculty Members.** We encourage you to discuss academic or personal questions with us during office hours or via email. These discussions need not be limited to CHE 20500 content.

K. **Academic Dishonesty.** Academic dishonesty ***will not be tolerated*** in any form in this course. Specifically, Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Section B-2-a, Code of Student Conduct] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972] All incidents of academic dishonesty will be reported to the Dean of Students. **Such incidents include: (i) possessing or accessing, in hardcopy or electronic form, solutions to the: course text, previous years' homework problems, and exams (e.g., obtaining solutions from websites such as Course Hero, Chegg, Quizlet, etc.); (ii) claiming credit for work that is not your own original work; (iii) enabling other students to create work that is not their original work; and (iv) collaborating with other students (or getting other outside help) during an exam. The punishment for the first offense is a grade of zero for the entire work (exam, quiz, or homework), and the punishment for a second offense is an F mark for the class.**

L. **Academic Integrity.** Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert your instructors and university officials to potential breeches of this value by either emailing integrity@purdue.edu or by calling (765) 494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

M. **Student-Initiated Purdue Honors Pledge.** As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.

N. **Student Professionalism.** The highest standards of professionalism and ethics are expected in CHE 20500. Each student is expected to come to class on time and not disrupt the class. Each student is expected to follow Purdue's student conduct code and behave in a professional manner. The rights of students in violation of the code of conduct are outlined on Purdue's website. Each student is expected to exhibit consideration and respect towards the other students, the teaching assistants (TAs), the graders, and the faculty members. Each student is expected to exhibit a positive attitude. Expectations for each student include (but are not limited to):

a. Attending all class sessions.
b. Coming to class on time and prepared by reading assigned material beforehand.
c. Refraining from disrupting class (e.g., turning off or silencing mobile phones, refraining from mobile phone or laptop use during class, and carrying on a loud conversation during class).
d. Maintaining the highest standards of academic honesty and integrity.
e. Being an active contributor to team assignments.
f. Being knowledgeable about the policies and information described in the syllabus.

O. **Key Course Dates.**

Lectures (in person):	Monday and Wednesday	9:30 a.m. – 11:20 a.m.	ARMS B061
Exam 1	Th 2/12	8:00p - 09:30p	UC 114
Exam 2	Th 3/5	8:00p - 09:30p	UC 114
Exam 3	W 4/15	8:00p - 09:30p	UC 114
Final Exam	To Be Announced, **As Set by the University**		

No Class	1/19	MLK Day
	3/16, 3/18	Spring Break
	3/23, 3/25, 4/15	in lieu of evening exams

Class Drop Deadline with a Withdrawal (W): April 12, 2026

P. **Emergency Procedure for the ARMS building.** In the event that the class would need to evacuate ARMS (e.g., in the event of a fire alarm), the class should proceed to exit the building and meet by the PUSH building. Do not leave the area as emergency responders will need to count to ensure that all persons have made it from the facility. In the event that we are required to shelter in place (e.g., due to a tornado warning), we will remain in ARMS B061 or gather in another suitable location in the basement level of ARMS.

Q. **Office Hours**

a. **Professor Hebner (**thebner@purdue.edu, Office: FRNY 2043A)
Office Hours (Location: FRNY 3062B): Monday, 2:30 pm – 4:00 pm

b. **Sooryadas Sudhakaran** (sudhak10@purdue.edu)
Office Hours (Location: FRNY 1142): Tuesday, 5:30 pm – 7:00 pm

c. **James Zheng** (zheng632@purdue.edu)
Office Hours (Location: FRNY 1142): Wednesday, 5:30 pm – 7:00 pm

R. **Website for Course Information (purdue.brightspace.com).** This course will use the Brightspace site. The website is limited to enrolled students and will have the syllabus, homework assignments, and other important class information associated with it. To login, use your university name and password. Please check the website regularly for assignments. Important announcements will be posted on the Brightspace page and will not be automatically sent to e-mail. To get an e-mail/text notification of new announcements, follow the directions here.

S. **E-mail.** Occasionally, important class announcements will be disseminated through the class e-mail list. It is your responsibility to regularly check your e-mail every day and to read the e-mails regarding CHE 20500 to receive important class information. E-mail is the preferred mode of contact. Please put CHE 205 in your subject line. If you e-mail the instructors with questions or a request to make an appointment, please allow a minimum of 24 hours for a response during the week or a response by Monday at 5 pm if the e-mail is sent on the weekend.

T. **Assessment of Course Outcomes.** A weighted average grade will be calculated as follows.
Concept Quizzes: 5%
Homework: 15%
Exams: 60% (3 x 20% each)
Final Examination: 20%

The grading scale will be as follows.

A range: 100 – 85% of the weighted points
B range: 84.9 – 75% of the weighted points
C range: 74.9 – 65% of the weighted points
D range: 64.9 – 55% of the weighted points
F: Less than 55% of the weighted points

The final course grades *may* be scaled to a higher value, as determined by the faculty instructor at the conclusion of the course. Grades will never be scaled downward from the distribution above. As a rule, it cannot be guaranteed that scaling will be applied and students should refer to the distribution above for information regarding overall course grades. There is no preset distribution of final grades. The grading will reflect demonstrated student capability relative to an absolute performance standard that is expected of all Purdue chemical engineering students, rather than a scale that compares students to a mean performance metric on any evaluation vehicle. In practice, this means the entire class could receive A marks.

U. **In-Class Quizzes**. There will be occasional in-class quizzes related to concepts from the reading or problem solving in class. You will be instructed whether the quiz should be completed individually or in groups. Each quiz will have equal weight, and **the two lowest quiz grades will be dropped.**

V. **Electronic Homework.** Electronic homework will be performed in groups and turned in via Gradescope. ***Begin each problem on a new sheet of paper, and number the pages***. The solution to each problem ***must*** include a picture or flow chart (hand or computer-generated) of the system or problem of interest, a listing of the known quantities and their units of measurement, and a listing of the unknown quantities that must be determined. ***If your solution to a problem does not contain these items, it will not be graded and you will receive a grade of 0 for that problem.*** The homework will be representative of content posed on the midterm and final examinations. As such, the purpose of the homework is to ensure that learners are comfortable with the course content. **Homework solutions will not be posted. It is your responsibility to get help either before the problems are due or after they have been submitted for grading.**

W. **Late Homework.** All assignments are due on the stated date and time given when the homework is assigned. **Late homework will not be accepted.**

X. **Homework Grading.** Homework will be graded on the basis of 10 points per problem. A problem worked perfectly or with 1 or 2 minor errors will get 10 points. A problem with more than 2 minor errors but no major (logic) errors will get 7.5 points. A problem with 1 major error or more than 3 minor errors will get 5 points. A problem with more than 1 major error will get 2.5 points. A problem with no credible effort will get zero points. The assessment of your performance and contribution by your teammates will be used to scale your homework grades. **Your lowest scaled homework score will be dropped**.

Y. **Team Evaluations.** An ability to function effectively on teams is a learning objective of this course. Students will be assigned to two different teams during the semester to complete homework assignments. The first team pairing will be for the first half of the assignments, and the second team pairing will be for the second half of the assignments. Students will use CATME to submit information used for Team Formation and Peer and Team Evaluations. At the halfway and end points of each team pairing, students will be required to log into CATME and rate their performance as well as the performances of their team members. Each student's point total for the team homework assignments completed in that timeframe will be multiplied by the multiplier to obtain the final point total for those homework assignments. The CATME software will use the evaluations to compute a "multiplier" with a value between 0.00 (very poor contributions to group) and 1.05 (extremely excellent contributions to group) for each group member. If one is interested in how these multipliers are calculated, please see the research papers posted at https://info.catme.org/research/. Note that multipliers of 0.00 are very rare; typically, multipliers are between 0.80 and 1.05.

Z. **Homework Cover Page.** Each homework assignment must have a cover sheet. A template of the cover sheet is posted on Brightspace (in the Homework module) as a fillable Microsoft Excel sheet. The cover sheet ***must contain*** the printed first and last names of the group members, the date, and the homework assignment number in the appropriate locations. Below this identifying information, the following statement MUST appear:

"Each signature below attests that the signer contributed significantly to the solution of all problems in this homework assignment".

All team members who contributed must sign and print their names next to the signature. The signature and printed names must be clearly legible. IF A TEAMMATE DID NOT PARTICIPATE IN THE SOLUTION OF THE HOMEWORK, THEN THIS TEAMMATE SHOULD NOT SIGN. If multiple groups worked together, indicate that on the cover page. Otherwise, identical solutions will be regarded as cheating. ***If this page is not present, the homework will be awarded a grade of zero and will be returned ungraded. If a team member does not participate in the solution of all the problems on the assignment, that team member will receive a grade of zero on the entire homework assignment.***

AA. **Examinations.** Timed examinations will be conducted in person. For each examination you will be supplied with one or more pages of relevant equations. You will not be allowed to use any books or notes in addition to these equations pages, which means that all you will be allowed to have on your desk during the examination period is: the exam itself, the notes pages provided, the paper on which you are writing solutions, something with which to write, and a calculator. All other electronic devices are forbidden, including cell phones and pagers. These must be turned off and may not be handled at any time during the exam. Students caught with other materials during an exam will be assumed to be cheating. Remember that no collaboration or outside help is allowed. The final exam is comprehensive (i.e., it will cover the entire contents of the course).

Any student who cannot take an exam as scheduled (e.g., religious holiday, conflicts with another exam) *must* make special arrangements by sending the instructors an e-mail at least one month *before* the exam is given. In cases of extenuating circumstances (e.g., illness, quarantine, bereavement) or extreme duress (e.g., hospitalization), please provide documentation to Prof. Hebner, and if possible, speak to one of the instructors *before* the exam takes place. Travel plans do not constitute extenuating circumstances.

BB. **Regrade Requests for Homework and Exams.** A student ***has one week after the graded exam or homework has been released*** to submit a regrade request via Gradescope. This is the only means by which to have work re-graded in this course. Any homework assignments or exams submitted for re-grading will be *re-graded in their entirety* and may be marked lower than the original score that was received. For exams, regrade requests will not be accepted until after the exam solutions have been posted and students have compared their solutions to the posted solutions.

CC. **Accessibility.** Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let us know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: (765) 494-1247. If you are a student with any form of individual learning needs, please speak with the professors whether or not you seek an accommodation so that we are aware of your circumstance and can deliver course content in a manner that is most compatible with your learning situation.

In addition to the University policy, the Davidson School of Chemical Engineering has established procedures for students seeking accommodations. These can be found online at the ChE Undergrad Office website. Only those accommodation requests that conform to both University and ChE policy guidelines will be implemented.

Some important points from the ChE policy include: Please give letters of accommodation to Prof. Hebner, and your academic advisor (and not the graduate TA for office hours). If you have your letter at the start of the term, we strongly recommend you give it to us within the first two weeks of the semester. If your accommodation involves exam conditions, we strongly urge you to provide a minimum of one week notice to ensure that the accommodations requested are available.

DD. Campus Emergencies. In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. ***Here are ways to obtain information about changes in this course. You are expected to check your @purdue.edu email address frequently.***
Course Brightspace Page: Spring 2026 CHE 20500-004 LEC
Instructors' email address: thebner@purdue.edu

EE. Nondiscrimination Statement. Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies and Statements.

FF. Attendance. This course follows the Academic Regulations: Attendance and Office of the Dean of Students: Class Absences policy posted in Brightspace under "University Policies and Statements." This policy states that students are expected to be present for every meeting of the classes in which they are enrolled. It is understood that, occasionally, you may miss lecture due to unforeseen circumstances (e.g., illness – please do not come to class if you are feeling ill), and, in these cases, you can ask any questions about missed concepts during office hours. Unless it falls under the University excused absence regulations (see below), any work due should be submitted on time. In cases falling under excused absence regulations, the student or the student's representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent medical care.

GG. Illness. If a student becomes sick (e.g., with flu-like symptoms), the student should seek prompt medical attention, and then not come back to class until the student has been symptom-free for more than 24 hours. That is, the student should utilize the resources at the Purdue University Student Health Center (PUSH) or another trained medical professional. Materials will be made available electronically to assist any students who are ill, and reasonable accommodations will be made on an individual basis to ensure that all students have the opportunity to learn. If possible, the student should let Professor Hebner know as soon as possible such that they can aid in bringing the student up to speed in the course material as rapidly as possible. In the event of a severe outbreak of illness at Purdue that mandates class not meet, all attempts will be made to deliver the course online.

HH. Mental Health/Wellnes Statement. Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765) 494-6995 during and after hours, on weekends and holidays, or through its counselors physically located in the Purdue University Student Health Center (PUSH) during business hours.

II. Basic Needs Security. Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 am - 5 pm Monday through Friday.

JJ. Use of Copyrighted Materials. Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Thus, these materials cannot be posted online (e.g., Chegg, Course Hero, etc.). Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally. Notes taken in class are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor. To obtain permission to sell or barter notes, the individual wishing to sell or barter the notes must be registered in the course or must be an approved visitor to the class. Course instructors may choose to grant or not grant such permission at their own discretion, and may require a review of the notes prior to their being sold or bartered. If they do grant such permission, they may revoke it at any time, if they so choose.

KK. Course Material as Listed by Topic.

1. Materials Balances
   a. Open and Closed Systems
   b. Steady-state Systems
   c. Systems in a Transient State
   d. Multi-unit Systems
   e. Balances on Reactive Systems
   f. Balances on Multiphase Systems
2. Solution Thermodynamics
   a. Raoult's and Henry's Laws
   b. Binary Vapor-Liquid Equilibrium
2. Solution Thermodynamics (Continued)
   c. Binary Liquid-Liquid Equilibrium
   d. Vapor-Liquid-Liquid Equilibrium
3. Energy Balances
   a. First Law of Thermodynamics
   b. Balances on Non-reactive Processes
   c. Balances on Reactive Processes
4. Combined Mass and Energy Balances in Systems in a Transient State

LL. Course Material as Listed by Text Chapter.

Chapters 2, and 3 Engineering Calculations
Chapter 4: Fundamentals of Material Balances
Chapter 5: Single Phase Systems
Chapter 6: Multiphase Systems
Chapter 7: Energy and Energy Balances
Chapter 8: Balances on Nonreactive Processes
Chapter 9: Balances on Reactive Processes
Chapter 10: Balances on Transient Processes

MM. Disclaimer. All information in this syllabus is subject to change. If any change occurs, it will be announced in the class and/or posted on Brightspace.